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Unraveling the emergence of collective behavior in networks of cognitive agents

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Nicola Zomer ✉ & Manlio De Domenico ✉

Cognitive agents, powered by Large Language Models (LLMs), possess advanced reasoning and communication capabilities that fundamentally distinguish them from non-cognitive particles, which rely solely on formal rules. While their ability to replicate human individual and social behaviors is still under scrutiny, the impact of their embedded "intelligence" on emergent behaviors remains poorly understood. Here, by comparing cognitive agents with classic particles, we investigate how LLM capabilities shape emergent phenomena in two tasks: function optimization and social organization emerging from the Schelling model of segregation. To this aim, we introduce LLM Agent Swarm Optimization (llmASO), where a swarm of interacting LLM agents acts as an optimizer. Our findings reveal that, while individual agents outperform particles in decision-making, their consensus tendencies and ability to exploit patterns can make them prone to premature convergence. Adjusting network topology can alleviate this effect, but typically at the cost of slower overall convergence compared to classical Particle Swarm Optimization (PSO). In contrast, in the Schelling model, we demonstrate that local interactions and homophilic mechanisms allow cognitive agents to generate distinct emergent behaviors, underscoring the importance of realistic communication architectures in complex social simulations. This work clarifies how LLM capabilities introduce new mechanisms for collective behavior and has implications for future applications of LLM agents in swarm robotics, social experiments, and complex decision-making tasks.

According to Russell and Norvig[1], a rational agent is one that "acts so as to maximize the expected value of a performance measure based on past experience and knowledge". Cognitive agents operate on the same principles, learning from past experience to perform goal-directed, optimized actions. This requires a range of mental processes, such as perception, memory, and reasoning, that allow them to adapt and react to complex environments[2,3].

These capabilities distinguish them from classic particles, which rely only on local rules and non-verbal interactions without any individual awareness or goal-directed behavior. While such particles can still display complex individual and group behaviors[4–8], the underlying mechanisms differ significantly from cognitive agents, raising questions about how cognition influences emergent dynamics in group settings.

Interestingly, language, and by extension large language models (LLMs), enable agents to perceive, plan, and interact in ways that resemble human cognition. Through textual inputs, language model agents can perceive their environment, engage in logical reasoning and planning, and even verbally interact with each other[9]. Together with knowledge and memory, these components form a proper "cognitive architecture" of the agent[10] and allow them to be utilized in place of human subjects in behavioral experiments and complex systems research[11–14].

However, LLM capabilities should not be confused with human cognition, as it is still unclear whether these models exhibit genuine intelligence. Although LLMs have passed several tests designed to assess it[15,16], new and more complex tests are constantly being developed, with some scientists even arguing that "any linguistic test is unlikely to tell us much about common sense or genuine intelligence"[17]. Therefore, although we use the terms LLM and cognitive agents interchangeably in this work, it is important to note that our focus remains exclusively on LLM systems.

In particular, some recent works have explored the emergence of consensus in LLM populations, showing how language model agents, through iterative interactions, can naturally form stable conventions and consensus without explicit coordination, echoing dynamics in human societies[18,19]. In addition, it is known, for example, that LLM agents tend to spontaneously converge towards opinions in line with scientific reality[20]. However, it remains unclear how this affects the dynamics of complex systems and the resulting emergent properties. In fact, with cognitive agents, intelligence is not only an emergent property of the system, as with particles,

Department of Physics and "Galileo Galilei", University of Padua, Padua, Italy. ✉e-mail: nicola.zomer@gmail.com; dmanlio.dedomenico@unipd.it

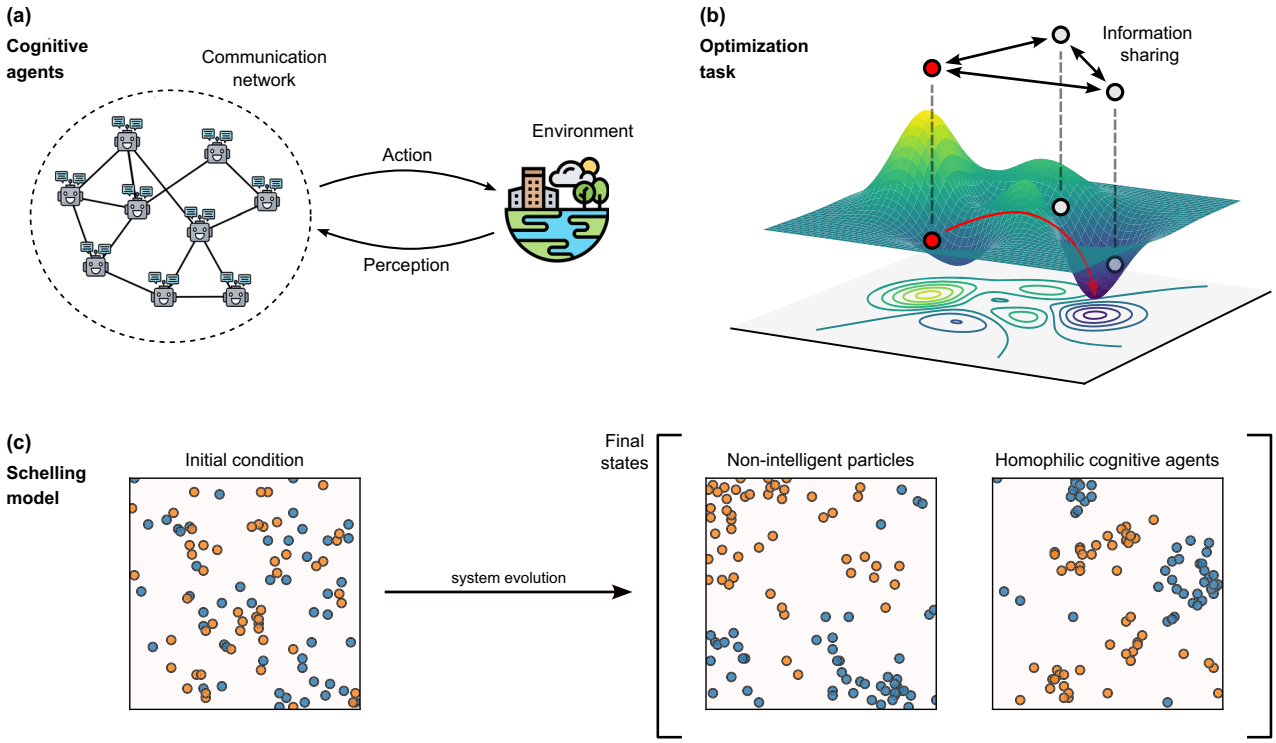

Fig. 1 | Research overview: illustration of the research carried out and the experiments performed. a Cognitive agents interact with the environment via natural language, receiving information and performing goal-oriented actions. Language capabilities also allow them to verbally interact with each other, according to a predefined communication network. Icons from Flaticon.com. **b** In function optimization, the exchange of suggestions and previous solutions allows agents to avoid local minima and correctly converge to the global optimum. **c** Cognition can alter the dynamics of well-known social experiments, such as the Schelling model of segregation.

but is also embedded within each entity, fundamentally altering the underlying dynamics.

Here, we investigate the effect of LLM capabilities on collective emergent phenomena with a particular focus on the role of information flow and spontaneous consensus in shaping system dynamics. This is achieved by comparing the dynamic behavior of LLM agents with classic particles on two distinct tasks: function optimization and a variant of the Schelling model of segregation[21]. Optimization is chosen as an example of a goal-oriented task requiring multi-agent coordination, motivated by recent work showing that a single LLM can be used as an optimizer in simple settings[22]. In contrast, the Schelling model serves as a well-established benchmark in the social sciences, illustrating how complex collective patterns can emerge from simple interactions[23]. Together, these two paradigms capture complementary aspects of collective behavior. A research overview is presented in Fig. 1.

Since cognitive agents somehow emulate aspects of human interaction, the results are also discussed in light of controlled experiments on human collectives. These include studies on collaborative learning in networks[24,25], social computation[26], the spread of behavior in online social networks[27], and the spontaneous emergence of social conventions[28].

Results

Three widely used benchmark functions, the Ackley, Rastrigin, and Rosenbrock functions, are considered for the optimization task (see Fig. 2a). The main results are obtained by testing different methods on all of them, in 2 dimensions, and in a bounded region characterized by $x_i \in [-10, 10]$ for $i = 1, 2$. The results include Optimization by PROmpting (OPRO)[22], a framework for utilizing a single LLM as an optimizer. Our method, LLM Agent Swarm Optimization (llmASO), extends this approach to a swarm of agents that share information and engage in multi-agent debate through a predefined communication network. Particle Swarm Optimization (PSO)[29], a widely used algorithm inspired by the social behavior of swarms, is chosen as a benchmark for particles as it closely resembles our method, differing primarily in how cognitive and social learning are implemented, as discussed in Methods. In particular, the Clerc and Kennedy variant (CF-PSO) is considered[30].

For the segregation dynamics, a variant of the classical Schelling model is explored where agents can move freely in space. As described in Methods, this reduces hallucinations, preventing the dynamics from collapsing into random motion. The system consists of 100 agents evenly distributed between two types, each being happy if the fraction of agents of the other type in its neighborhood is below a specified tolerance $T \in [0, 1]$. For simplicity, only a few tolerance levels are studied in this work:

$$T_{set} = \{1/10, 2/10, \ldots, 8/10, 9/10\}\,. \tag{1}$$

Unlike particles, which are always randomly relocated, agents can exploit their past experiences and neighborhood information to make informed decisions. In addition, each agent engages in a direct interaction with another before relocating, allowing them to exchange information and potentially collaborate[11].

Cognitive agents and the exploration-exploitation trade-off

Figure 2b shows a comparison between llmASO, OPRO, CF-PSO, and a single particle version of PSO. The comparison is performed in terms of the number of function evaluations, with a maximum of 9000. For swarm-based systems (llmASO and CF-PSO), 20 independent runs are performed, whereas for single-agent systems such as OPRO, only 5 runs are considered. For comparison reasons, the initialization of agent positions is kept the same across the models, but changes from run to run. For both CF-PSO and llmASO, the swarm size is set to 20.

The horizontal convergence lines for the single particle reflect its inability to learn and improve its location autonomously. In contrast, cognitive agents can independently identify optimal paths in simple or unimodal optimization landscapes. This stems from their ability to identify

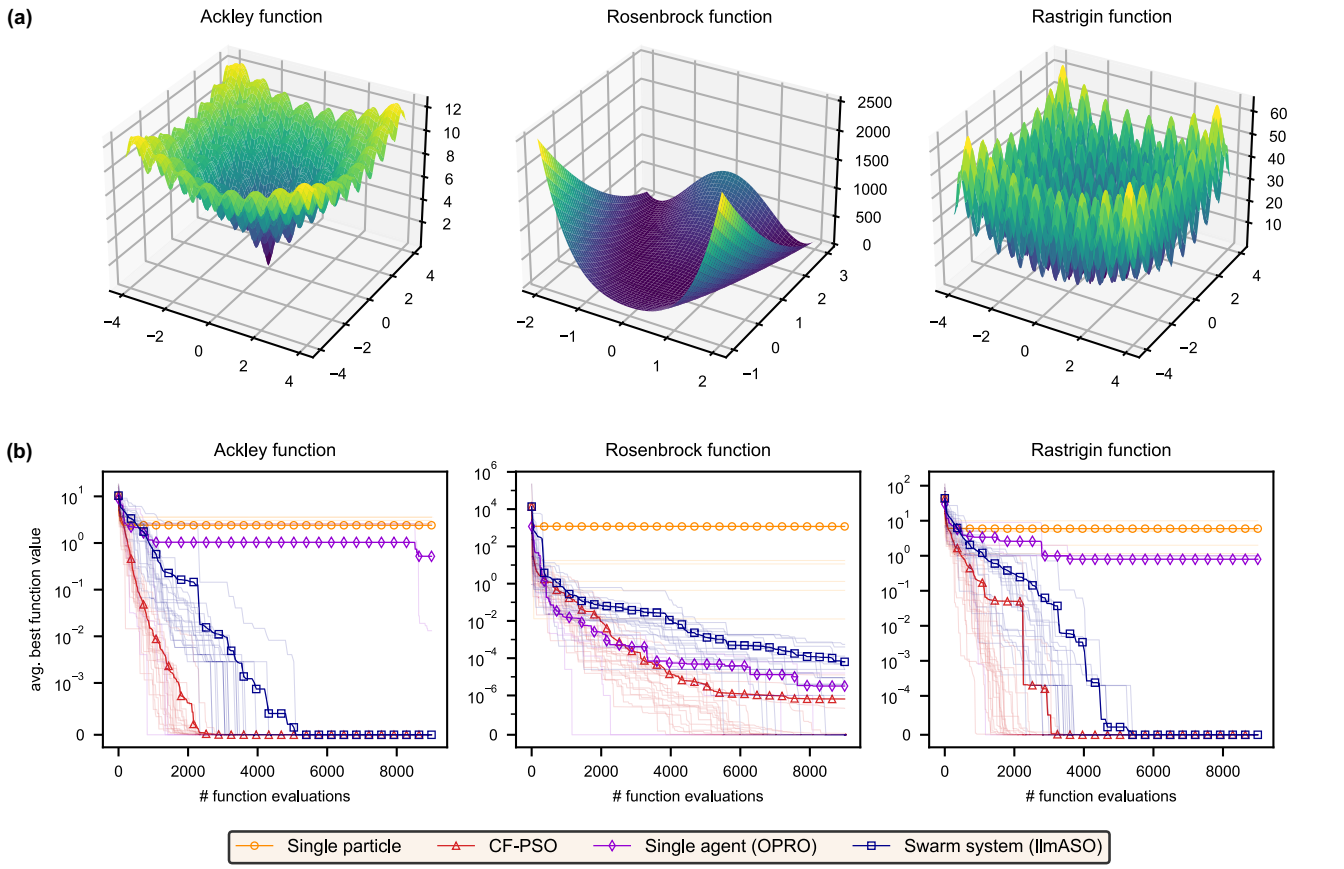

Fig. 2 | Benchmark functions considered and convergence results. a Close-up visualization of the considered optimization landscapes near the global optimum. Notice the strong presence of local minima in Ackley and Rastrigin, while a narrow parabolic valley characterizes the optimum region in the Rosenbrock function. **b** Due to LLM capabilities, single agents (OPRO) clearly outperform isolated particles. In contrast, llmASO converges more slowly than CF-PSO, but still finds the global minimum in all cases without requiring a mathematical formulation of the optimization problem. Number of runs: 20 for CF-PSO and llmASO, 5 for a single particle and OPRO.

regularities across previous solutions and extract patterns, unlike particles that simply move toward the current best. As a result, OPRO often exhibits rapid convergence to the global minimum. For instance, in the Ackley function, OPRO reaches the global best in 3 out of 5 runs, each requiring fewer than 1500 function evaluations, while in Rosenbrock it converges faster than llmASO, matching the performance of CF-PSO, albeit over a smaller number of runs.

However, optimization landscapes with many local minima, such as Ackley and Rastrigin, require multi-agent collaboration for consistent convergence, as the exploitation capabilities of single agents make them vulnerable to premature convergence when working alone. Unlike OPRO, llmASO consistently reaches the global optimum in these cases. This observation aligns with literature on human behavior, which shows that collectives outperform individuals in complex problems due to the diffusion of effective solutions through the network[25]. Similarly, the tendency of human individuals to copy or explore near their own best solutions is reflected in single agents, as OPRO often gets trapped in local minima in multi-modal landscapes. In addition, llmASO reaches a final average swarm diameter of approximately 2 across all benchmark functions, significantly smaller than the search space dimension of 20. This suggests that agents stop extensive exploration of the search space, with global convergence spontaneously emerging from local interactions, effectively forming a universally shared social convention[28].

Compared with CF-PSO, in the studied setting, llmASO converges more slowly but retains the advantage of not requiring a mathematical formulation of the problem. This slower convergence can be attributed to our deliberate choice of using a ring network topology in llmASO. However, this is essential for cognitive agents: in a fully connected network, the small population size would quickly lead to consensus formation toward the current local best[19,20]. Instead, limiting the information flow through a ring topology preserves agent diversity, enabling minority groups to establish alternative exploration paths that may eventually overcome dominant behaviors[18]. We confirmed this in preliminary tests, where topologies with smaller average shortest path lengths, such as Barabási–Albert (BA) or random geometric graphs (RGGs), led to convergence failures on the Rastrigin function in approximately 50% of the trials, as shown in Supplementary Information (SI), Fig. S2. While this result aligns with previous research indicating that inefficient networks outperform efficient ones in complex environments, with efficiency defined by how quickly information spreads[31], contrasting findings have been observed in human experiments[25]. This difference can be attributed to the rapid emergence of consensus in small networks of LLM agents[19] and to their inherent consistency during multi-agent debate[32], both leading efficient networks to favor exploitation. In addition, the observed problem-dependence of the optimal network structure, with BA dominating in Rosenbrock, is another well-known phenomenon in collaborative learning[24].

Emergence of segregation

Schelling model outcomes are quantitatively compared using the following metrics, averaged over multiple realizations $\langle\cdot\rangle$:

- the segregation coefficient[33], defined as:

$$s = \frac{2}{N_{ag}^2} \sum_{\{c\}} n_c^2 \,, \tag{2}$$

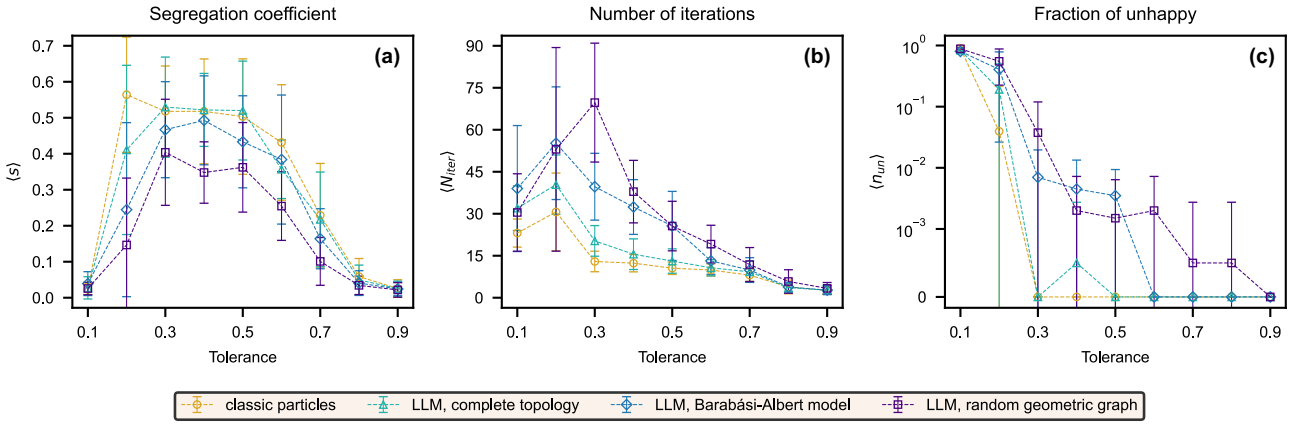

Fig. 3 | Impact of LLM capabilities and network topology on the segregation model dynamics. a Average segregation coefficient $\langle s \rangle$, **b** average number of iterations $\langle N_{\text{iter}} \rangle$, and **c** average fraction of unhappy agents $\langle n_{\text{un}} \rangle$ as a function of the tolerance T. Error bars indicate ±1 standard deviation across simulations. Classic particles and fully connected agents show nearly identical results, confirming the robustness of the Schelling model. When considering a BA or RGG topology, a decrease in the segregation coefficient and an increase in the number of iterations are observed. A step-like decrease instead appears in the fraction of unhappy agents. Number of simulations: 20.

where $\{c\}$ is a set of clusters, each of size n_c, and N_{ag} is the total number of agents (see SI for details);

- the number of iterations N_{iter} required to reach a stationary state (capped at 100);
- the final fraction of unhappy agents n_{un}.Notice that $s \to 1$ for complete segregation, i.e., only one cluster per type is formed, while $s \to 0$ when the number of clusters approaches N_{ag}.

The first results are presented in Fig. 3, showing the phase transitions of the above metrics for classic particles and cognitive agents. Each simulation is performed 20 times, starting from the same initial configurations for comparison purposes. Since the communication network plays a central role in shaping the interactions between cognitive agents, different topologies are studied in this case, starting with a complete network where all agents are connected. Then, the BA model is chosen for its ability to well approximate many real-world systems and is randomly initialized at the beginning of each run and then kept fixed. Finally, the RGG is selected to study the impact of local interactions. For this topology, the radius is set equal to that defining the neighborhood (i.e., 15), so that the resulting communication network is consistent with the physical properties of the system.

First, the 3 classic phases of the Schelling model can be identified at different tolerance levels in all cases[33]. The mixed phase, occurring at very low T, is characterized by agents moving dynamically in space without forming clusters or improving overall satisfaction, resulting in $\langle s \rangle \approx 0$, a high $\langle N_{iter} \rangle$, and $\langle n_{un} \rangle \approx 1$ (Fig. 3a, b, c, respectively). As T increases, the system transitions toward a diluted segregated state, characterized by a final state where almost all agents are happy, and clusters are present, i.e., $\langle n_{un} \rangle \approx 0$ and $\langle s \rangle \gg 0$. Then, for high T, the system undergoes another transition and a frozen state emerges, where the static nature is caused by the fact that happy agents are not moved, and already in the initial configuration, many are happy. Consequently, $\langle s \rangle \approx 0$, reflecting the initial state, $\langle N_{iter} \rangle$ is small as the system rapidly reaches a stationary configuration, and $\langle n_{un} \rangle \approx 0$ due to the high T. The continuous nature of this transition reflects the results of the original Schelling model with a high density of vacancies[33].

The Schelling model demonstrated robustness to the addition of LLM capabilities and verbal interactions under a complete network topology. In this case, the large population size and the high network connectivity result in a low interaction rate between the same pair of agents, complicating the formation of consensus dynamics. This is consistent with large-scale experimental evidence that behaviors spread more effectively in clustered-lattice networks than in random ones[27]. As a result, the segregation coefficient of agents remains unchanged relative to particles for all T, even in terms of fluctuations. The same pattern is observed for $\langle N_{iter} \rangle$ and $\langle n_{un} \rangle$, with only a slight increase in the first for agents. However, a difference appears when considering the average number of neighbors in the final states $\langle S_{nn} \rangle$: as shown in Fig. 4, in the mixed and diluted phases, cognitive agents consistently reach a higher $\langle S_{nn} \rangle$ than particles.

When considering a BA topology, the lower connectivity combined with the presence of hubs favors the formation of smaller clusters, causing $\langle s \rangle$ to decrease, while increasing $\langle N_{iter} \rangle$ and introducing a step-like decreasing behavior in $\langle n_{un} \rangle$. This becomes even more evident with RGG, where agents are limited to local interactions, making them more inclined to explore their immediate neighborhood, spread behaviors, and form local clusters characterized by long-term interactions. Still, the mixed, segregated, and diluted phases persist across all network topologies, with transitions occurring at similar values of T.

Homophily in segregation dynamics

Homophily, i.e., the tendency of individuals to associate with similar others, has been observed in a variety of social networks[34]. While the Schelling model bases its dynamics on this attribute, the inclusion of verbal interactions allows us to study the impact of its addition to the communication network. To achieve this, we incorporate homophily into the BA model following Karimi et al.[35], where the probability that a node j connects to node i is defined as:

$$\Pi_i = \frac{h_{ij} k_i}{\sum_l h_{lj} k_l}, \tag{3}$$

where k_i is the degree of node i and h_{ij} is the homophily between nodes i and j:

$$h_{ij} = \begin{cases} h, & \text{if i and j are of the same type} \\ 1 - h, & \text{otherwise} \end{cases}. \tag{4}$$

In this work, we focus only on extreme scenarios, with agents only interacting with others of the opposite or same type, respectively showing complete heterophilic ($h = 0$) and homophilic ($h = 1$) behavior. These are compared with the default scenario, represented by the classic BA model, where no preference is made between the types (homogeneous case, $h = 0.5$).

Homophily allows cognitive agents to escape the mixed phase, which disappears from the phase diagram in Fig. 5a, whereas heterophily shifts the transition towards higher tolerances. This result highlights that network structure can strongly shape system dynamics, determining whether segregation emerges, in line with previous findings on the formation of social norms in large populations[28]. Moreover, homophily causes the system to reach a stationary state as quickly as particles, while in the heterophilic case,

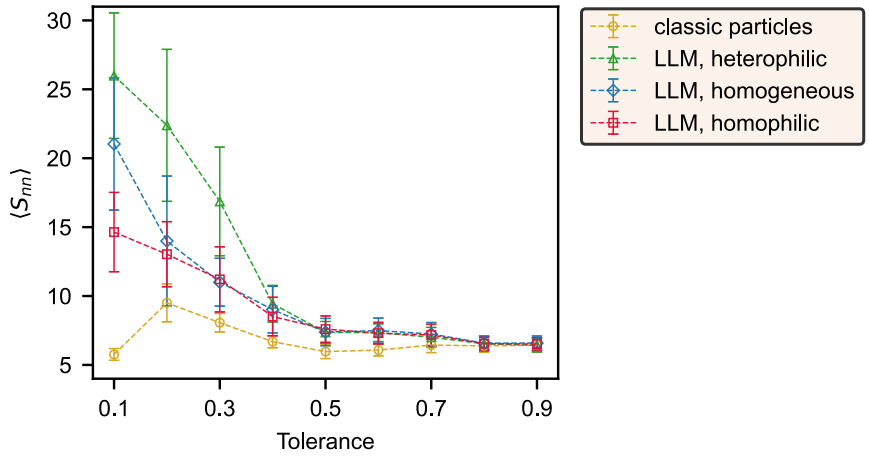

Fig. 4 | Average nearest neighbor size. Long-range verbal interactions bring cognitive agents closer together, increasing $\langle S_{nn} \rangle$ compared to particles. Heterophilic interactions amplify this phenomenon by bringing together many agents of different types.

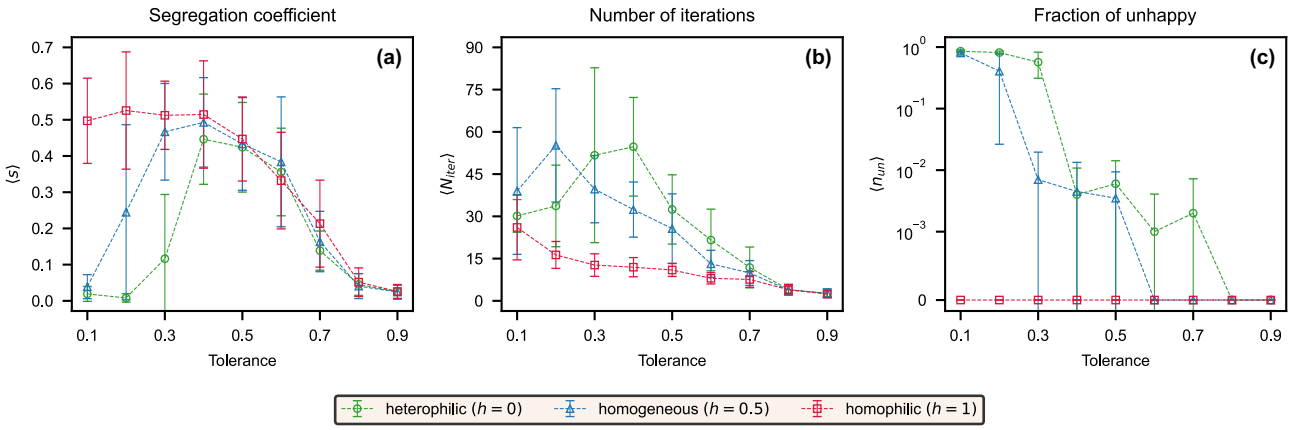

Fig. 5 | Effect of network homophily on the Schelling model. a Average segregation coefficient $\langle s \rangle$, **b** average number of iterations $\langle N_{\text{iter}} \rangle$, and **c** average fraction of unhappy agents $\langle n_{\text{un}} \rangle$ as a function of the tolerance T. Error bars indicate ± 1 standard deviation across simulations. Homophilic interactions facilitate cluster formation and bring complete happiness ($n_{un} = 0$) across all runs and tolerance levels. Homophilic agents can escape the mixed phase, while heterophily shifts the mixed-segregated transition towards higher tolerances. This behavior is absent in classic particles, which cannot establish homophilic or heterophilic interaction networks. Number of simulations: 20.

many more iterations are required, as shown in Fig. 5b. This is expected, since interactions among agents of the same type facilitate cluster formation, both in terms of cooperative movements and information sharing. In contrast, heterophily counteracts this, as pairs of opposite types struggle to cooperate effectively, and while agents continue to observe the location and state of all their neighbors, they collect broader system-level information only from agents of the other type. Still, the high value of $\langle s \rangle$ in the heterophilic case for $T \in [0.4, 0.6]$ is an important and non-intuitive result since it shows that, for intermediate tolerances, strong segregation emerges even when agents are forced to always interact with those of the other type.

Observe also that homophily brings all agents to happiness in every run, across all tolerance levels, as shown in Fig. 5c, where $\langle n_{un} \rangle$ is always zero with no fluctuations when $h = 1$. In contrast, for $h = 0$, this metric still exhibits a step-like decrease, similar to the homogeneous case, but the curve shifts toward higher tolerances, reflecting the difficulty of reaching global happiness in the heterophilic setting.

Finally, Fig. 4 shows that the average number of neighbors increases significantly in the heterophilic case. This occurs because heterophilic interactions bring many agents of different types into proximity, forming dense clusters of unhappy agents that cause the emergence of a frozen state even for higher tolerances.

Discussion

The research and analysis presented in this study reveal that the strong individual abilities of cognitive agents do not automatically translate into superior performance over classical particles in complex tasks that require multi-agent collaboration, such as the optimization of multi-modal functions. Similarly, in the Schelling model under a fully connected network topology, no clear advantage in reaching happiness is observed for cognitive agents, whose behavior remains nearly identical to that of particles in the studied setting. Nevertheless, both experiments show a clear dependence of the results on the communication network, echoing previous findings on how network structure influences exploration and exploitation in problem-solving[24], as well as social coordination in human groups[26].

Specifically, in complex problems requiring extensive exploration, we found that the consensus tendency of LLM agents, combined with their ability to exploit patterns, can increase the risk of premature convergence to a local best. This can be understood through the lens of consensus dynamics: when collaborating, small groups of homogeneous agents often reach a single opinion too quickly[19], frequently aligning with factual information[20], avoiding adequate exploration of the search space. To address this, we introduced the network topology in llmASO as a hyperparameter controlling the exploration-exploitation trade-off, showing that a ring topology leads to stable optimization over the Ackley, Rosenbrock, and Rastrigin functions. However, this adjustment reduces system efficiency, with llmASO converging more slowly than CF-PSO across all functions, despite the inherent superiority of individual agents. Interestingly, these results contrast with Mason and Watts' experiments on human collectives, which show that efficient networks can improve problem-solving performance and reduce the tendency of individuals to copy each other[25]. Together, these

observations highlight the importance of further investigating the role of network structure in consensus dynamics and multi-agent debate, as well as of providing guidelines for the design of communication architectures for multi-agent systems with language capabilities, such as robot swarms[36].

Another possible solution to mitigate the premature convergence of llmASO could be to introduce an explicit mechanism to balance individual and social information, similarly to how PSO uses acceleration coefficients[29]. In fact, over-emphasizing either source can promote premature convergence: excessive weight on individual information encourages agents to act independently, reducing the system to multiple parallel executions of OPRO, while excessive weight on social information can cause the system to collapse into a single collective trajectory. In the current llmASO formulation, prompts present individual information before social information, as detailed in the SI. Because later information tends to have a stronger influence on LLM outputs[37], this ordering may implicitly over-weight social cues, leading to a phenomenon known as *herding*[38]. Possible strategies to control this effect include explicitly prompting the agents to equally weight their own information and others' suggestions, or running OPRO and the debate independently and combining the generated candidates.

Our results with the Schelling model show that the choice of the network topology is crucial in social experiments as well. In our experiments, this not only enabled more realistic simulations, as suggested by Ashery et al.[18] and motivated by Iacopini et al.[39], but also played a key role in generating dynamics different from those of classic particles. In fact, the Schelling model proved robust to the addition of language model capabilities in a complete topology, both due to its intrinsic resilience to various realistic refinements[40,41] and because of the large size of the system, which limits the exploitation of cognitive abilities for optimal positioning and complicates the formation of long-term interactions and consensus dynamics. Differently, when local interactions and homophilic mechanisms are considered, cognitive agents can produce distinct dynamics compared to particles, leading to emergent states with reduced segregation.

This result motivates future work aimed at extending our model by incorporating additional variables, which could potentially foster long-term collaborations and produce new emergent behaviors. For example, the rate of interactions could be modeled by drawing inspiration from human interactions, reflecting a priority based on the memory of previous interactions[42]. Similarly, aging (also called "social inertia") could be incorporated, making the rate of interactions depend on the persistence time of an agent in a state[43]. The agent architecture could also be modified to introduce long-term effects and memory of interactions, achieved by generating plans, incorporating self-reflections[44], or performing summarizing steps[11]. Finally, memory decay could be implemented with more complex strategies than a simple cutoff, such as adopting a power law or exponential kernel[45].

It is also important to note that our results were obtained using an experimental setup that, as is often the case in LLM research, depends on several parameters. While rigorous testing has been conducted on most of the models' hyperparameters by varying each individually while keeping others fixed (see SI), their combined influence on system dynamics, as well as their dependence on the optimization landscape, remain largely untested. In addition, factors such as LLM choice, prompt design, interaction model, agent architecture, and incentive structure could further alter the system behavior. For instance, studies of humans embedded in networks have shown that incentive structures strongly affect coordination and optimization outcomes[26], suggesting that similar effects could emerge in LLM systems. Therefore, future work is needed to generalize our findings across different experimental settings and system properties.

In this vast space of possibilities, our work should be seen as a first step toward a better understanding of the impact of "cognitive" capabilities on collective behaviors. Researchers and developers could benefit from it to design better experiments or applications, both in swarm robotics and social sciences. For instance, our insights on modularity in task-oriented settings could be applied to scenarios requiring extensive exploration, such as search and rescue operations, disaster management, or border control. Alternatively, carefully designed communication networks could be implemented in LLM-based social experiments to more accurately represent societies, potentially improving our understanding of their dynamics. Finally, LLM-based experiments could inspire new mathematical models formalizing the strategies observed in LLM agents. For instance, in the Schelling experiments, particles could be modeled to interact, exchange neighborhood information, and move toward positions of higher expected happiness according to specific rules.

Methods

Optimization: background and compared algorithms

One of the first attempts to use LLMs as optimizers has been made by researchers at Google DeepMind[22], who proposed a new method (OPRO) based on the idea of replacing the mathematical formulation of the optimization procedure with a problem description in natural language. Their method simply consists of using a single LLM to iteratively generate new solutions from a prompt containing the 20 best previously generated solution-value pairs, sorted in descending order, thus not considering the possibility of creating a multi-agent system. This brings many limitations, including sensitivity to initial conditions and convergence issues.

To overcome them and perform a better comparison with non-intelligent particles, we propose a new method (llmASO) based on the definition of a network of agents that, at each step, simultaneously share suggestions and solution-value pairs before generating new candidate solutions. The resulting information flow allows for enhancing mathematical and strategic reasoning via multi-agent debate[46], which promotes consistency across multiple model generations[32]. In addition, convergence is facilitated through the exchange of a set of best previous solutions. This approach offers an alternative method for tuning the exploration-exploitation trade-off, rather than relying only on the LLM temperature as in OPRO.

This is a huge advantage since playing with the LLM temperature carries the risk of getting stuck on the same instruction for many steps if the value is too low, or aggressively exploring the search space, even ignoring the previous trajectory if it is too high[22]. Instead, a multi-agent system can favor exploration by maintaining a stronger differentiation between its entities and enhance exploitation by increasing the information flow or network connectivity. A complete description of our model, including the hyperparameter values used in the experiments, is available in SI, Section 1.

In the case of particles, information flow is limited to the exchange of previous solutions found, whereas the generation of new candidates must be explicitly defined through a set of rules exploiting this exchanged information. This is precisely the mechanism adopted in PSO[29], where the best individual position p_d^i and the best position found by the whole swarm g_d along all dimensions d are used to update particle i velocity and position according to the following rules:

$$v_d^i(t+1) = w(t)v_d^i(t) + c_1 r_{1,d}\left[p_d^i(t) - x_d^i(t)\right] + \\ + c_2 r_{2,d}\left[g_d(t) - x_d^i(t)\right], \tag{5}$$

$$x_d^i(t+1) = x_d^i(t) + v_d^i(t+1), \tag{6}$$

where c_1, $c_2 > 0$ are the cognitive and social acceleration coefficients, respectively, $r_{1,d}, r_{2,d} \sim \mathcal{U}(0,1)$, and w is a parameter introduced by Shi and Eberhart[47] to balance exploration and exploitation. Note that p_d^i and g_d effectively serve as cognitive and social learning signals, respectively.

Due to this resemblance and its wide application in complex optimization problems, we chose PSO as a benchmark method for non-intelligent particles, focusing on the Clerc and Kennedy variant for its simplicity and ability to find near-optimal solutions efficiently[30]. This variant modifies the previous equation by introducing the constraint factor (CF-PSO) in the

velocity update as follows:

$$v_d^i(t+1) = \chi \cdot \left[v_d^i(t) + c_1 r_{1,d}\left(p_d^i(t) - x_d^i(t)\right) + c_2 r_{2,d}\left(g_d(t) - x_d^i(t)\right)\right], \quad (7)$$

where $\chi = 2/\left|2 - \phi - \sqrt{\phi^2 - 4\phi}\right|$, $\phi = c_1 + c_2$, and $\phi > 4$ for convergence to be guaranteed. In the experiments, we set $c_1 = c_2 = 2.05$ as indicated by Clerc and Kennedy[30], resulting in $\phi = 4.1$ and $\chi = 0.729$.

Definition of the optimization problems

Three common benchmark functions are considered in this study: the Ackley, Rosenbrock, and Rastrigin functions. The first is chosen because of the presence of both a clear optimal direction and many local minima, the second because it is characterized by a narrow parabolic valley that makes convergence to the minimum difficult, and the third because of the large number of regularly distributed local minima that make the task extremely complex.

As already introduced, the main results are obtained by testing all functions in 2 dimensions, with $x_i \in [-10, 10]$ for all $i = 1, 2$. Agent locations are initialized uniformly in the search space, and the analytic form of the functions is not provided to them. Furthermore, experimentally, it is observed that cognitive agents are inclined to test the origin even without having data to support this decision, thus invalidating the results when the global best is in (0, …, 0). The same applies to the center of the allowed region when the domain boundaries are provided in the prompt. To avoid these biases, we randomly shift each function in every run while keeping the global optimum in a reachable location.

Finally, when working with cognitive agents, float numbers must be approximated to fit into the context window of the LLM. Unless otherwise specified, we set a precision value $\epsilon = 3$ in the experiments, such that optimizing a 2-dimensional function in $[-10, 10]^2$ becomes equivalent to finding the optimal location in a 20,000 × 20,000 grid.

Segregation model

In this work, we consider a variant of the original Schelling model[21], characterized by a very high density of vacancies. This choice was made after experimentally observing that, with few available positions as in the original model, agents tend to hallucinate the state they will find in each of them and select the best corresponding one. In contrast, when moving in an almost continuous space, agents are not constrained to a limited set of choices and can always exploit their collected information, which drastically reduces hallucinations.

In detail, we define a 100 × 100 grid and consider a total number of agents, N_{ag}, equal to 100, equally distributed between the 2 types and initially positioned randomly in space to form a mixed configuration. The high presence of vacancies requires introducing a radius so that two agents are neighbors only if their spatial distance is below a given threshold. This neighborhood radius is set to 15, such that the average number of neighbors in the initial configuration is 6.14 ± 0.41 (averaged over 10,000 random initialization), similar to the original Schelling model with a relatively low value of vacancy density. Notice that too low a radius would make any interesting dynamics disappear, since each agent would only have a couple of neighbors or none at all, while too large a value would generate highly populated neighborhoods, leading to complete segregation regardless of the underlying dynamics and causing problems in terms of context windows.

Furthermore, the dynamics is unconstrained, so unhappy agents are not forced to move only to satisfactory positions, and long-term relocations can occur. The movements are long-range, i.e., agents can move to any free location in the grid, and non-periodic boundary conditions are applied. Happy agents are not moved. In fact, also relocating them would slow down convergence to a stationary state[48], increasing the number of iterations and thus making the simulation too expensive in terms of LLM calls.

The system evolves until one of the following stopping conditions is reached:

- all agents are happy;
- the dynamics reaches a stationary state;
- a maximum number of iterations, counted in Monte-Carlo steps, of 100 is achieved.The second condition is monitored by computing the segregation measure and the density of unhappy agents at each iteration and checking whether these values are stable over a predefined time window $\Delta\tau$. Stability is assessed by linearly interpolating the values of each metric over the time window and checking whether the absolute value of the associated slope is less than a given metric-dependent threshold (resp. Θ_s and Θ_{un}). The choice of $\Delta\tau$, Θ_s, and Θ_{un} is entirely experimental. Starting from some experiments with standard particles, the parameters are varied a posteriori until values are found that allow early stopping without affecting the final values of the selected metrics. In the results reported here, we used $\Delta\tau = 20$ and $\Theta_s = \Theta_{un} = 0.001$.

Finally, notice that particles and agents follow the same model, differing only in how they are relocated during the simulation. While particles are relocated by random motion, cognitive agents autonomously decide their next location after verbally interacting with a randomly selected connected peer.

System prompts

In both tasks, the system prompt consists of a short and fixed task description and a list of previously explored solutions paired with their relative score. Cognitive agents are then instructed to leverage the collected information to generate new solutions, without explicitly defining an update rule or providing example strategies. Such solutions must adhere to given constraints, also described in natural language. Finally, to avoid vague and generic answers, agents are asked to end their responses with the new location.

For llmASO, only the best previous solutions are provided, avoiding duplicates and sorted with the best values at the bottom. The ordering is motivated by Yang et al., who demonstrated that this sorting improves accuracy and accelerates convergence in optimization tasks[22], as LLM outputs are more influenced by instructions at the end of the prompt[37]. Here, agents are also prompted to think step-by-step, adopting the Take a Deep Breath prompting technique[22].

For Schelling, the prompt also includes a description of the current state, i.e., the agent's location, state, and neighborhood. Simulating local perception, the latter consists of a list of the location and type of all neighbors, arranged in decreasing order of distance, and limited to a maximum of 20 agents to account for context window length. For the same reason, and because too old states are irrelevant, a memory cutoff is implemented, ensuring that only location-state pairs from the last 100 steps are provided. To reduce hallucinations, Schelling agents are provided with the possible transitions they can make, described in natural language (see SI for details). For the same reason, any prompting technique for step-by-step reasoning is avoided. In fact, we experimentally found that ending the prompt with "Let's think step by step" significantly increases the rate of hallucinations here, contrary to what is observed in the optimization task.

The exact system prompts used in the simulations are provided in the SI.

Agents interactions

Optimization agents interact synchronously, meaning that agents connected by the network communicate simultaneously, without following any specific order. This is implemented by leveraging the concept of a multi-agent debate, which has been shown to enhance mathematical and strategic reasoning while improving factual accuracy by contrasting hallucinations[32,46]. In particular, before the movement is performed, suggestions and a predefined number of the best previous solutions are shared through the network.

Differently, Schelling agents engage in agent-to-agent dialog, inspired by the work of Park et al.[11]. After the system prompt, each agent is provided with the type and state of the other. Once the conversation is initiated, agents

are given the dialog history and asked to generate a message to convey to their peer. The message has to be enclosed within the keys |<speak>| and |</speak>| to facilitate parsing. Additionally, the current location and state of the encountered agent are summarized and stored for future decisions. A distinction is also made between whether the "helper" agent is happy or not. If so, his role is solely to inform the other agent's decision, while, if not, he also participates in the relocation. In the last case, agents can move together, collaborate, or even swap positions, effectively performing a bilateral interaction.

More details on the interaction models and associated prompts are provided in the SI.

Parsing LLM output

In this work, we avoid constraining the agent's output solely to the new location using JSON formatting or grammars[49]. In fact, while answers obtained this way are simple to parse, they are typically synthetic, preventing the generation of complex reasoning, which is critical for step-by-step reasoning or agent communication. Instead, we allow the agent's response to remain free of format constraints and use the language model itself to extract the desired new location.

Implementation details

All algorithms are built using LangChain[50], a framework for developing language model applications. The experiments are run with Starling-LM-7B-alpha[51] (in 4-bit quantization format), an open-source, 7 billion parameter LLM developed by researchers at Berkley that delivered the best performance in preliminary studies compared to other models of the same size, such as Neural-Chat-v3.3[52] and OpenChat-3.5[53]. The main motivation for this choice is that we are interested in the practical application of these simulations to real-world swarms of small robots, each equipped with a local LLM instance, which imposes stringent requirements in terms of memory and computational power[36]. To run the models locally, we rely on the *llama.cpp* library[49] and its Python bindings *llama-cpp-python*[54].

Data availability

There is no primary data associated with this study. The generated and analyzed output data are provided as supplementary files.

Code availability

To ensure reproducibility of the experiments, all code used in this work will be made publicly available upon acceptance at https://github.com/CoMuNeLab/LLM-Agents. This includes the implementations of llmASO and the Schelling model with cognitive agents.

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Author contributions

N.Z. conducted the primary research, performed the analyses, and drafted the manuscript. M.D.D. supervised the project, provided guidance throughout the research process, and contributed to the interpretation of the results. All authors have read and approved the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Nicola Zomer or Manlio De Domenico.

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